

Chapter 1

Piecewise Functions

Piecewise functions are a kind of break from the composite and inverse functions we've been working on up to this point. Piecewise functions are best used when the formula deciding the input changes after a certain x value. Examples often used with piecewise functions include tax brackets, shipment/delivery price changes, discounts dependent on age, etc.

Before hopping into any examples, I think it's best to get acquainted with what a piecewise function looks like and how to graph it. Consider the below piecewise function:

$$f(x) = \begin{cases} x & 0 \leq x \leq 4 \\ -(x - 4) + 4 & 4 < x \leq 8 \\ \frac{x-8}{2} & x > 8 \end{cases}$$

What do you think this looks like? I've written the explanation of this on the next page, so take a moment to think about what happens before moving on.

- Why are there compound inequalities next to equations?
- Why are some of the inequalities $<$ and others $\leq$?
- What's up with that $x > 8$ in the bottom row? Why is it different?
- Could these compound inequalities be written in some other way?

1.1 Describing Piecewise Functions

Alright. Here it is.

The $f(x)$ is the name of the function, as is the case with any other function. It is set equal to a curly bracket containing cases.

Each row first identifies an equation and next by an interval that the equation applies within.

Using the above function as an example:

$$x \qquad 0 \leq x \leq 4$$

means to say that the equation $y = x$ applies for all x values between 0 and 4.

$$-(x - 4) + 4 \qquad 4 < x \leq 8$$

means to say that the equation $y = -(x - 4) + 4$ applies for all x values between 4 and 8 excluding 4. The value $x = 4$ is excluded because it's already accounted for in the previous domain.

$$\frac{x - 8}{2} \qquad x > 8$$

means to say that the equation $\frac{x-8}{2}$ applies for all x values that are greater than 8. Again, the inequality is not $x \geq 8$ because the value $x = 8$ was accounted for in the previous domain.

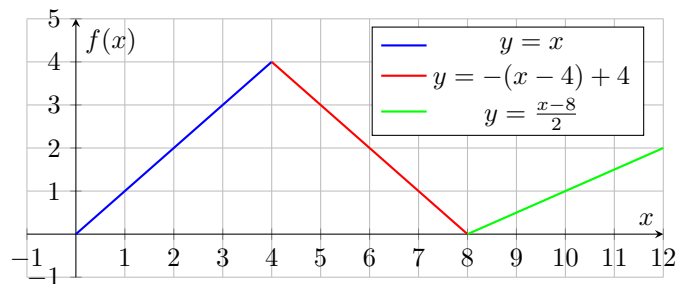
The inequality $x > 8$ could be written as $8 < x < \infty$ so that it matches the other inequalities.

The piecewise function could also be rewritten using interval notation, like so:

$$f(x) = \begin{cases} x & x \in [0, 4] \\ -(x - 4) + 4 & x \in (4, 8] \\ \frac{x-8}{2} & x \in (8, \infty) \end{cases}$$

This piecewise function produces the graph below:

Figure 1.1: Graph Of Piecewise Function $f(x)$



1.2 Example with Piecewise Functions

Now that we know what a piecewise function *is*, let's look at an example of where it may be used.

"Totally real bank name" gives money out for free. If we say that x represents the number of hours that have passed, then:

- The first 4 hours, represented as $0 \leq x \leq 4$, \$10 are given out per hour.
- The following 9 hours, represented as $4 < x \leq 13$, \$7 are given per hour.
- Every hour after that, represented as $x > 13$ or $13 < x < \infty$, \$5 are given out per hour.

Instead of writing each and every case, we can use a piecewise function to set all three cases equal to a function f . This function will output the amount of money given out after x hours.

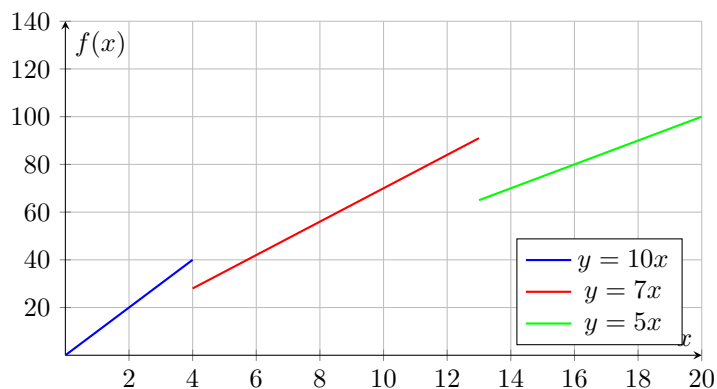
1.2.1 Potential Trap

We should be careful when constructing such a function, however. We can't simply call the equations $10x$, $7x$ and $5x$.

$$f(x) = \begin{cases} 10x & x \in [0, 4] \\ 7x & x \in (4, 13] \\ 5x & x \in (13, \infty) \end{cases} \quad \leftarrow \text{Bad Piecewise Function}$$

Doing so creates the piecewise function above and graph in Figure 1.2 below.

Figure 1.2: Piecewise Function with Incorrect Equations



Just throwing these into a piecewise function would create a valid function, but one that doesn't represent the situation given. For it to represent the provided situation, the lines need to be connected. Let's examine why.

If 4 hours passed, and 10 dollars were given out every hour, then 40 dollars were given out.

When the fifth hour passes, the piecewise function moves to the next equation, which we've currently assigned as $7x$. Plugging $x = 5$ to $7x$ gives $7 \times 5 = 35$. That's 5 dollars less than what was given out in the first 4 hours.

Since the amount given out in the fifth hour must add to the amount already given on the first 4, we expect the amount after 5 hours (i.e. $f(5)$) to be 47 dollars.

You may think, "just shift the function up 12 units ($y = 7x + 12$). Doing that gets us up to the 47 we expect ($35 + 12 = 47$).” Although this *would* work, it requires that you know how much was given out at the fourth hour.

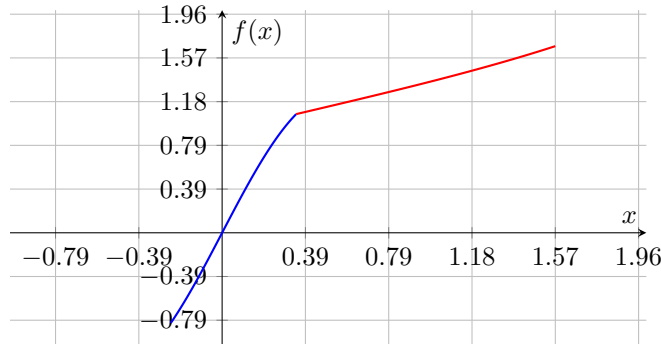
Of course you can do that in this example. It’s easy. But something like the one below would not be very easy to determine:

- $\sin(\pi x) + \tan\left(\frac{x}{2}\right)$ acts on $x \in \left[-\frac{\pi}{13}, \frac{\pi}{9}\right]$
- $\tan\left(\frac{\pi^2 x}{22}\right)$ acts on $x \in \left(\frac{\pi}{9}, \frac{\pi}{2}\right]$

The solution to that looks like this if you’re curious:

$$g(x) = \begin{cases} \sin(\pi x) + \tan\left(\frac{x}{2}\right) & x \in \left[-\frac{\pi}{13}, \frac{\pi}{9}\right] \\ \tan\left(\frac{\pi^2(x - \frac{\pi}{9})}{22}\right) + \sin\left(\frac{\pi^2}{9}\right) + \tan\left(\frac{\pi}{18}\right) & x \in \left(\frac{\pi}{9}, \frac{\pi}{2}\right] \end{cases}$$

Figure 1.3: Super Hard Problem (Scale: $\frac{\pi}{8}$)



1.2.2 The Correct Way To Do It

Let's go back to the problem we were working on.

Instead of predicting or calculating how much we should add or subtract, we can literally move the entire function using vertical and horizontal transformations.

In this case, we know that $y = 10x$ up until $x = 4$.

We're already given that $x = 4$, and $10 \times 4 = 40$, meaning $y = 40$.

What's left is to apply those transformations (4 right and 40 up) to $7x$.

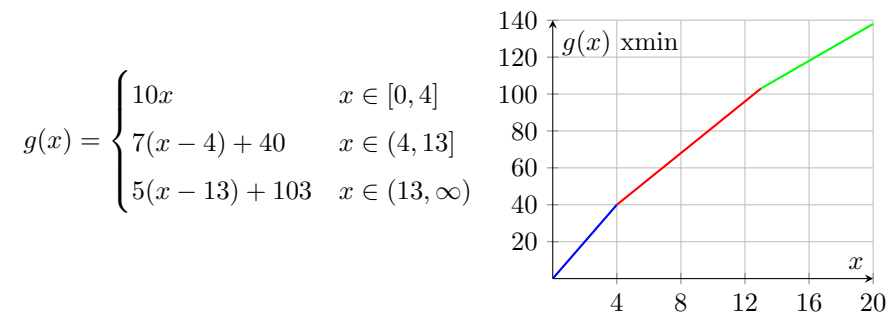
That gives: $y = 7(x - 4) + 40$.

This method does not rely on any calculations. All you need is the right-most edge of the bound (4 in this case) and the value of the equation at that bound (40 in this case). The same can be done for the last function.

In the first 4 hours, the bank gave out 40 dollars. In the next 9 hours, the bank gave out $7 \times 9 = 63$ dollars. That means the total amount given is $40 + 63 = 103$ dollars.

We also know that the right-most boundary of the second equation is $x = 13$. That means we need to transform $5x$ 13 units to the right and 103 units up, giving $y = 5(x - 13) + 103$.

Putting that all together gives the piecewise function and graph below:



1.3 Jump Discontinuity

Up to this point, we've been working with functions that continue cleanly from one another. To illustrate this point, I'll use another example. For those of you also taking physics, this sort of example may look familiar:

Problem

Janet is a frequent runner and takes the same route to her friend's house every day. On her daily runs, she:

- Runs 1000 m in 30 minutes.
- Rests for 5 minutes.
- Runs 750 m in 30 minutes.
- Rests for 10 minutes.
- Runs 1200 m in 45 minutes.

Create a piecewise function and a Velocity vs. Time graph (VT Graph) representing Janet's velocity.

For those who haven't taken physics, a Velocity vs. Time graph is just like a normal function but the x -axis is time t and the y -axis is velocity v .

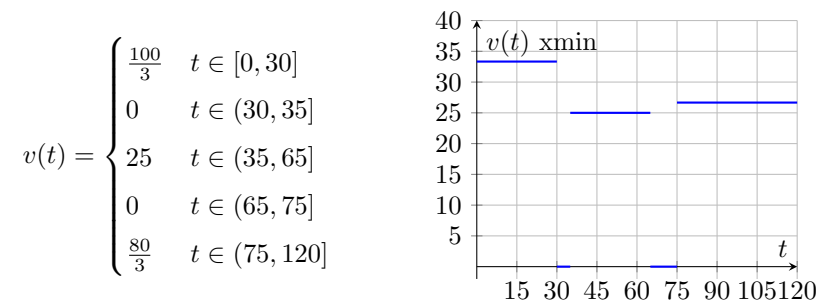
Also, instead of using $f(x)$ or $g(x)$, I'll be using $v(t)$, a velocity function with time (in minutes) as the independent variable.

We're not given any velocities, but we can find them quickly. Velocity is calculated by dividing the distance traveled by the time it took to travel that distance. For example, if you drove 30 miles in 2 hours then you were going $30\text{ mi}/2\text{ hr} = 15\text{ mi/hr}$.

Using the equation $v = \frac{\Delta s}{\Delta t}$ (s stands for change in position btw), we get:

$$\begin{array}{ccccc} v_1 = \frac{1000}{30} & v_2 = \frac{0}{5} & v_3 = \frac{750}{30} & v_4 = \frac{0}{10} & v_5 = \frac{1200}{45} \\ v_1 = \frac{100}{3} & v_2 = 0 & v_3 = 25 & v_4 = 0 & v_5 = \frac{80}{3} \end{array}$$

We don't have any variables here, so there's nothing to line up. That means we can create the piecewise function and graph right now:

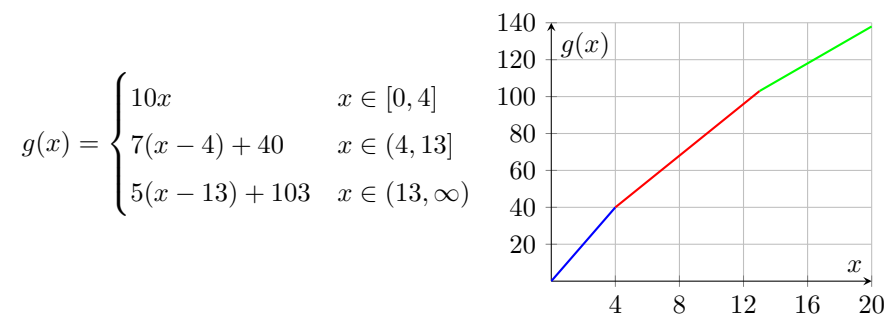


This function doesn't smoothly transition like with the bank example. It breaks off and continues in different places. It jumps from $v = 35$ all the way down to $v = 0$, hence why it's called a "jump" discontinuity. This is **not** continuous. Also... 25 minutes per minute? Dayum she's slow. But that's beside the point.

1.4 Removable Discontinuity

A removable discontinuity is a less obvious than a jump discontinuity. The function may *look* continuous, but it'll have a hole somewhere. I won't spend too much time on this because it's solved almost identically to regular piecewise functions. In fact, I can actually use a previous example to demonstrate this.

Here was the result from that bank example:



This function is continuous because there are no gaps. But what if we changed the inequality $x \in [0, 4]$ to $x \in [0, 4)$?

If we did that, the function would begin at 0, continue all the way up to $3.99999\dots$, then skip 4 to go to $4.00000\dots 1$ and continue as normal. Making some substitutions would help visualize what's going on.

$$f(3.9) = 39$$

$$f(4) = \text{undefined}$$

$$f(4.1) = 40.7$$

Since the function has nothing defined for $x = 4$ (i.e. it's skipped), it's labeled as undefined. The function f has no value at $x = 4$. On a graph, that is usually labeled by an open circle.

With that change, the piecewise function and graph now looks like this:

